

Detim Zhao

✉ detimzhao@gmail.com

🌐 [linkedin.com/detimzhao](https://www.linkedin.com/detimzhao)

🐙 github.com/DetimZhao

Education

Bachelor of Science, Software Engineering

May 2026

Arizona State University, Polytechnic – GPA: 3.88 / 4.00

Mesa, AZ

- Barrett (Honors), Dean's List (Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2025)

Technical Skills

Languages – Java, Python, C, C++, PostgreSQL, MySQL, Bash/Shell Scripting, UML

Technologies – pandas, scikit-learn, matplotlib, seaborn, NumPy, PyTorch, Transformers (Hugging Face), OpenCV

Tools – Docker, AWS, Git, GitHub, Unreal Engine 5 (UE5), LaTeX, Jupyter Notebook, PlantUML, Astah, Photoshop, Illustrator, Excel

Methodologies – Scrum, Agile Development, Git Workflows, Branching Strategies

Experience

Extended Reality (XR) Creative Developer

July 2024 – Present

Meteor Studio at ASU

Mesa, AZ

- Developing and researching a proprietary volumetric rendering pipeline using OpenCV for camera calibration and alignment, while collaborating with a team to innovate and solve technical challenges. Co-author of two research posters to present ideas at HotMobile 2025. (Python)
- Implemented automated scripts to capture images/video using a kinetic camera and process calibration data, reducing a previously manual week-long process to an efficient, automated workflow. (Python, Bash/Shell)
- Developed a research prototype in Unreal Engine 5 (UE5) to simulate a stereoscopic-camera-equipped drone autonomously following a runner on a track, utilizing a custom UI for fine-tuning motion control.

Undergraduate Researcher

June 2023 – Aug 2023

Texas A&M-Corpus Christi

Corpus Christi, TX

- Partnered to design and develop a Python program to operate multiple PX4 drones in a gazebo simulation through terminal controls, achieving successful control of up to 8 to 10 drones. (Python)
- Preprocessed two datasets, then trained and tested five ML models (KNN, SVM, Random Forest, Naive Bayes, Ensemble) for gesture recognition, enhancing classification accuracy by 10%. (Python)
- Co-authored a research paper for publication (DOI: [10.1145/3565287.3617621](https://doi.org/10.1145/3565287.3617621)) that documents findings and work, leading to publication in ACM and conference presentations. (LaTeX)

Projects

Clustering Analysis of Steam Games Using Reviews, Sentiments, and Metadata

- Conducted clustering analysis on 42,497 Steam game records using K-Means and Agglomerative Clustering, implementing custom distance metrics (binary for categorical, Euclidean for numerical) to optimize performance, and documented the process and findings in an academic paper. (Python)
- Scraped and analyzed the sentiment of 385 games' English reviews using VADER and vectorized text with BERT embeddings, transforming Steamworks API JSON data into structured formats for clustering and analysis.
- Engineered a modular and reproducible Python workflow for end-to-end data collection, preprocessing, and analysis using pandas, scikit-learn, and transformers, with version control via Git and Git LFS for scalability and efficiency.

Console-Based Chat Application

- Designed and Organized local host pSQL database to store information about the user and messages, ensuring efficient data storage and queries. (Java, PostgreSQL)
- Developed console-based directions and options for users to register, log in/out, update account info, create a chat room, and etc., streamlining user interactions.
- Implemented commands when the user is in a chat room to display active users, chat history, and more, improving overall user experience.

Maze Generator and Solver

- Developed a GUI-based maze generator and solver where the user specifies the dimensions of the maze, improving user experience and usability. (Java, JavaFX)
- Implemented a recursive backtracking algorithm to solve the maze and draw the solution path, successfully solving and visually representing maze solutions with the GUI.